

EECS 182/282A Final Exam Practice Questions
RMS→RMS Norms, μ P, and the Muon Optimizer

20 Comprehensive Questions with Solutions

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Instructions

This exam contains 20 questions covering material from Lectures 6-7. Show all work for partial credit. Clearly state any assumptions you make. A horizontal line separates each question from its answer.

1 Question 1 (10 points)

Consider a linear layer with weight matrix $W \in \mathbb{R}^{512 \times 2048}$ mapping from dimension $d_{\text{in}} = 2048$ to $d_{\text{out}} = 512$.

(a) (4 pts) If the spectral norm of W is $\|W\|_2 = 4.0$, what is $\|W\|_{\text{RMS} \rightarrow \text{RMS}}$?

(b) (6 pts) Suppose you want to constrain $\|\Delta W\|_{\text{RMS} \rightarrow \text{RMS}} \leq 0.1$ for an update. What is the maximum allowable spectral norm $\|\Delta W\|_2$? Explain why this constraint gives different effective learning rates to layers with different dimensions.

ANSWER:

(a) Using the relationship between RMS→RMS and spectral norms:

$$\|W\|_{\text{RMS} \rightarrow \text{RMS}} = \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}} \|W\|_2 = \sqrt{\frac{2048}{512}} \cdot 4.0 = \sqrt{4} \cdot 4.0 = 2 \cdot 4.0 = 8.0$$

(b) Rearranging the relationship:

$$\|\Delta W\|_2 = \|\Delta W\|_{\text{RMS} \rightarrow \text{RMS}} \cdot \sqrt{\frac{d_{\text{out}}}{d_{\text{in}}}} = 0.1 \cdot \sqrt{\frac{512}{2048}} = 0.1 \cdot \sqrt{\frac{1}{4}} = 0.1 \cdot 0.5 = 0.05$$

Explanation: With a single global RMS→RMS constraint $\gamma = 0.1$, different layers automatically get different spectral norm budgets through the $\sqrt{d_{\text{out}}/d_{\text{in}}}$ factor. A layer with larger fan-in gets a smaller spectral budget, effectively implementing layer-wise learning rate scaling without manual tuning. This is the key insight for width-aware optimization.

2 Question 2 (8 points)

(a) (4 pts) Define what it means for a rectangular matrix $Q \in \mathbb{R}^{m \times n}$ to be semi-orthogonal. Give the condition for both cases: $m \leq n$ and $m > n$.

(b) (4 pts) If $G = U\Sigma V^T$ is the SVD of a gradient matrix, show that UV^T is semi-orthogonal and state its condition number on its support.

ANSWER:

(a) A matrix $Q \in \mathbb{R}^{m \times n}$ is semi-orthogonal if:

- When $m \leq n$: $QQ^T = I_m$ (rows are orthonormal)
- When $m > n$: $Q^TQ = I_n$ (columns are orthonormal)

(b) For UV^T where $U \in \mathbb{R}^{m \times r}$ and $V \in \mathbb{R}^{n \times r}$ have orthonormal columns (from SVD):

If $m \leq n$:

$$(UV^T)(UV^T)^T = UV^T VU^T = U(V^T V)U^T = UU^T = I_m$$

If $m > n$:

$$(UV^T)^T(UV^T) = VU^T UV^T = V(U^T U)V^T = VV^T = I_n$$

Therefore UV^T is semi-orthogonal. Since all non-zero singular values equal 1, the condition number is:

$$\kappa(UV^T) = \frac{\sigma_{\max}}{\sigma_{\min}} = \frac{1}{1} = 1$$

This is optimal - the update has uniform strength in all directions.

3 Question 3 (12 points)

Consider the constrained optimization problem:

$$\min_{\|\Delta W\|_2 \leq \eta} \langle G, \Delta W \rangle$$

where $G = \nabla_W \mathcal{L}(W)$ is the gradient.

(a) (6 pts) Using the SVD $G = U\Sigma V^T$, derive the optimal solution ΔW^* . You may use the fact that the inner product is maximized (most negative) when vectors are aligned.

(b) (6 pts) Explain geometrically why $\Delta W^* = -\eta \cdot UV^T$ rather than $\Delta W^* = -\eta \cdot U\Sigma V^T / \|G\|_2$.

ANSWER:

(a) We want to minimize $\langle G, \Delta W \rangle$ subject to $\|\Delta W\|_2 \leq \eta$.

By Cauchy-Schwarz inequality:

$$|\langle G, \Delta W \rangle| \leq \|G\|_2 \cdot \|\Delta W\|_2$$

The most negative value is achieved when ΔW is aligned with $-G$ in the direction of the top singular vectors:

$$\langle G, \Delta W \rangle \geq -\|G\|_2 \cdot \|\Delta W\|_2$$

Using the budget fully with $\|\Delta W\|_2 = \eta$:

$$\Delta W^* = -\eta \cdot \frac{G}{\|G\|_2}$$

Since $\|G\|_2 = \sigma_{\max}$ and $G = U\Sigma V^T$:

$$\Delta W^* = -\eta \cdot \frac{U\Sigma V^T}{\sigma_{\max}} = -\eta \cdot U \cdot \frac{\Sigma}{\sigma_{\max}} \cdot V^T$$

For rank-1 case (or taking top singular direction): $\Sigma/\sigma_{\max} \rightarrow \text{identity on support}$, giving:

$$\Delta W^* = -\eta \cdot UV^T$$

(b) Geometric interpretation:

The solution $-\eta \cdot UV^T$ uses the full spectral budget η but distributes it **equally** across all principal directions (all singular values $\rightarrow 1$). In contrast, $-\eta \cdot U\Sigma V^T / \|G\|_2$ would preserve the anisotropy of the gradient - the largest singular direction would still dominate.

The UV^T direction removes the “elongation” caused by different singular values, creating an isotropic (condition number 1) update. This prevents the optimizer from being too sensitive to the gradient’s conditioning and makes steps more uniform across all relevant directions.

4 Question 4 (10 points)

For a batch size of 1, consider a linear layer $h = Wx$ where $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$.

(a) (4 pts) Show that $\nabla_W \ell$ is rank-1 by writing it explicitly as an outer product.

(b) (6 pts) For a rank-1 matrix A , prove that $\|A\|_2 = \|A\|_F$. (Hint: use the relationship between singular values and Frobenius norm.)

ANSWER:

(a) By the chain rule:

$$\nabla_W \ell = \frac{\partial \ell}{\partial h} \cdot \frac{\partial h}{\partial W}$$

Since $h_i = \sum_j W_{ij} x_j$, we have $\frac{\partial h_i}{\partial W_{ij}} = x_j$.

Therefore:

$$(\nabla_W \ell)_{ij} = \frac{\partial \ell}{\partial h_i} \cdot x_j$$

This can be written as:

$$\boxed{\nabla_W \ell = \frac{\partial \ell}{\partial h} \cdot x^T}$$

This is the outer product of vector $\frac{\partial \ell}{\partial h} \in \mathbb{R}^{d_{\text{out}}}$ with vector $x \in \mathbb{R}^{d_{\text{in}}}$, which has rank 1.

(b) For a rank-1 matrix A , the SVD has only one non-zero singular value: $A = \sigma_1 u_1 v_1^T$.

Spectral norm:

$$\|A\|_2 = \sigma_1$$

Frobenius norm:

$$\|A\|_F = \sqrt{\sum_{i,j} A_{ij}^2} = \sqrt{\sum_k \sigma_k^2} = \sqrt{\sigma_1^2 + 0 + 0 + \dots} = \sigma_1$$

Therefore:

$$\boxed{\|A\|_2 = \|A\|_F = \sigma_1}$$

5 Question 5 (12 points)

Consider signSGD with update rule $W_{t+1} = W_t - \eta \cdot \text{sgn}(G_t)$ where G_t is a rank-1 gradient at batch size 1.

- (a) (5 pts) Let $S = \text{sgn}(G)$ where each entry is ± 1 . Compute $\|S\|_F$ in terms of d_{in} and d_{out} .
- (b) (4 pts) Using the result from part (a) and the rank-1 property, find $\|S\|_2$.
- (c) (3 pts) Derive the constraint on η such that $\|\Delta W\|_{\text{RMS} \rightarrow \text{RMS}} \leq \gamma$ for some fixed γ .

ANSWER:

- (a) Since S has all entries equal to ± 1 :

$$\|S\|_F^2 = \sum_{i=1}^{d_{\text{out}}} \sum_{j=1}^{d_{\text{in}}} S_{ij}^2 = \sum_{i=1}^{d_{\text{out}}} \sum_{j=1}^{d_{\text{in}}} 1 = d_{\text{out}} \cdot d_{\text{in}}$$

Therefore:

$$\boxed{\|S\|_F = \sqrt{d_{\text{out}} \cdot d_{\text{in}}}}$$

- (b) Since G is rank-1, $S = \text{sgn}(G)$ preserves the rank-1 structure (up to sign patterns). For rank-1 matrices:

$$\|S\|_2 = \|S\|_F$$

Therefore:

$$\boxed{\|S\|_2 = \sqrt{d_{\text{out}} \cdot d_{\text{in}}}}$$

- (c) For the update $\Delta W = \eta S$:

$$\begin{aligned} \|\Delta W\|_{\text{RMS} \rightarrow \text{RMS}} &= \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}} \|\Delta W\|_2 = \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}} \cdot \eta \|S\|_2 \\ &= \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}} \cdot \eta \cdot \sqrt{d_{\text{out}} \cdot d_{\text{in}}} = \eta \cdot d_{\text{in}} \end{aligned}$$

To satisfy $\|\Delta W\|_{\text{RMS} \rightarrow \text{RMS}} \leq \gamma$:

$$\eta \cdot d_{\text{in}} \leq \gamma$$

$$\boxed{\eta \leq \frac{\gamma}{d_{\text{in}}}}$$

This shows signSGD requires learning rate to scale inversely with fan-in for stable RMS→RMS updates.

6 Question 6 (8 points)

Explain why the constraint $\|\Delta W\|_{\text{RMS} \rightarrow \text{RMS}} \leq \gamma$ applied to signSGD at batch size 1 becomes **conservative** (i.e., overly restrictive) when the batch size increases.

Your answer should reference:

- The relationship between rank and norms
- How batch size affects gradient rank
- The general inequality between spectral and Frobenius norms

ANSWER:

At batch size 1, the gradient G is rank-1, so $\|G\|_2 = \|G\|_F$ (equality holds).

As batch size increases, gradients are averaged over multiple samples:

$$G_{\text{batch}} = \frac{1}{B} \sum_{i=1}^B G^{(i)}$$

This creates a **higher-rank** matrix since we're summing rank-1 outer products from different samples.

For general matrices (rank ≥ 1), we have the **strict inequality**:

$$\|A\|_2 < \|A\|_F$$

The spectral norm $\|A\|_2 = \sigma_{\max}$ only captures the largest singular value, while the Frobenius norm $\|A\|_F = \sqrt{\sum_k \sigma_k^2}$ captures all singular values.

In signSGD analysis, we used $\|S\|_2 = \|S\|_F$ to derive the constraint. At higher batch sizes, this equality breaks down:

$$\|S\|_2 < \|S\|_F = \sqrt{d_{\text{out}} \cdot d_{\text{in}}}$$

Using $\|S\|_F$ as an upper bound for $\|S\|_2$ makes the constraint:

$$\eta \leq \frac{\gamma}{d_{\text{in}}}$$

overly conservative because we're overestimating the actual spectral norm. A tighter constraint could allow larger η at higher batch sizes by using better approximations to the true spectral norm (e.g., via Newton-Schulz orthogonalization).

7 Question 7 (10 points)

Consider the “scalar warm-up” analogy for P. Let $z_1, z_2, \dots, z_n$ be i.i.d. random variables with mean 0 and variance 1.

(a) (6 pts) Consider three scaling schemes for the sum:

- $A_n = \sum_{i=1}^n z_i$
- $B_n = \frac{1}{n} \sum_{i=1}^n z_i$
- $C_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n z_i$

What happens to each as $n \rightarrow \infty$? Which one has a stable, non-trivial limit?

(b) (4 pts) Relate this to neural network scaling: if you have a layer computing $h = Wx$ where $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, which scaling regime corresponds to keeping $\|h\|_{\text{RMS}}$ stable as $d_{\text{in}} \rightarrow \infty$?

ANSWER:

(a)

Scheme A: $A_n = \sum_{i=1}^n z_i$

- $\mathbb{E}[A_n] = 0$
- $\text{Var}(A_n) = n$
- As $n \rightarrow \infty$: A_n **blows up** (variance grows without bound)

Scheme B: $B_n = \frac{1}{n} \sum_{i=1}^n z_i$

- $\mathbb{E}[B_n] = 0$
- $\text{Var}(B_n) = \frac{1}{n}$
- As $n \rightarrow \infty$: $B_n \rightarrow 0$ **vanishes** (by Law of Large Numbers)

Scheme C: $C_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n z_i$

- $\mathbb{E}[C_n] = 0$
- $\text{Var}(C_n) = 1$
- As $n \rightarrow \infty$: $C_n \Rightarrow \mathcal{N}(0, 1)$ **stable non-trivial limit** (by Central Limit Theorem)

Only scheme C has a stable, meaningful limit distribution.

(b) For $h_i = \sum_{j=1}^{d_{\text{in}}} W_{ij} x_j$, this is analogous to summing d_{in} terms.

To keep $\|h\|_{\text{RMS}} = \frac{1}{\sqrt{d_{\text{out}}}} \|h\|_2$ stable:

We want each component $h_i \sim O(1)$, so:

$$\mathbb{E}[h_i^2] = \sum_{j=1}^{d_{\text{in}}} \mathbb{E}[W_{ij}^2 x_j^2] \sim O(1)$$

If $\mathbb{E}[x_j^2] \sim O(1)$, then we need:

$$\mathbb{E}[W_{ij}^2] \sim O(1/d_{\text{in}})$$

This corresponds to **scheme C**: scaling weights like $1/\sqrt{d_{\text{in}}}$ (Xavier/He initialization style). This is the regime where widths can scale while keeping activations stable - the foundation of P.

8 Question 8 (12 points)

State the two main desiderata for Maximal Update Parameterization (P) and provide a sufficient condition for achieving them.

(a) (4 pts) Write the two desiderata mathematically for layer ℓ .

(b) (4 pts) Write the sufficient condition on W_ℓ and ΔW_ℓ .

(c) (4 pts) Sketch why controlling RMS→RMS norms of weights and updates leads to stable activation updates. You may use the equation $\Delta h_\ell = \Delta W_\ell h_{\ell-1} + W_\ell \Delta h_{\ell-1}$.

ANSWER:

(a) The two P desiderata are:

(D1)	$\ h_\ell\ _{\text{RMS}} = \Theta(1)$	(activations stay stable)
(D2)	$\ \Delta h_\ell\ _{\text{RMS}} = \Theta(1)$	(updates stay stable)

(b) A sufficient condition is:

$\ W_\ell\ _{\text{RMS} \rightarrow \text{RMS}} = \Theta(1)$ $\ \Delta W_\ell\ _{\text{RMS} \rightarrow \text{RMS}} = \Theta(1)$

(c) Starting from:

$$\Delta h_\ell = \Delta W_\ell h_{\ell-1} + W_\ell \Delta h_{\ell-1}$$

Taking norms and using the triangle inequality:

$$\|\Delta h_\ell\|_2 \leq \|\Delta W_\ell h_{\ell-1}\|_2 + \|W_\ell \Delta h_{\ell-1}\|_2$$

By submultiplicativity of operator norms:

$$\|\Delta h_\ell\|_2 \leq \|\Delta W_\ell\|_2 \|h_{\ell-1}\|_2 + \|W_\ell\|_2 \|\Delta h_{\ell-1}\|_2$$

Converting to RMS norms (dividing by $\sqrt{d_\ell}$ on left, using operator norm relationships):

$$\|\Delta h_\ell\|_{\text{RMS}} \lesssim \|\Delta W_\ell\|_{\text{RMS} \rightarrow \text{RMS}} \cdot \|h_{\ell-1}\|_{\text{RMS}} + \|W_\ell\|_{\text{RMS} \rightarrow \text{RMS}} \cdot \|\Delta h_{\ell-1}\|_{\text{RMS}}$$

If all four terms on the right are $\Theta(1)$:

- $\|\Delta W_\ell\|_{\text{RMS} \rightarrow \text{RMS}} = \Theta(1)$
- $\|h_{\ell-1}\|_{\text{RMS}} = \Theta(1)$ (by D1 inductively)
- $\|W_\ell\|_{\text{RMS} \rightarrow \text{RMS}} = \Theta(1)$
- $\|\Delta h_{\ell-1}\|_{\text{RMS}} = \Theta(1)$ (by D2 inductively)

Then $\|\Delta h_\ell\|_{\text{RMS}} = \Theta(1)$, and the recursion remains stable across all layers and widths.

9 Question 9 (10 points)

In full Shampoo, we maintain left and right accumulators:

$$L_t = L_{t-1} + G_t G_t^T, \quad R_t = R_{t-1} + G_t^T G_t$$

and apply the update:

$$W_{t+1} = W_t - \eta \cdot L_t^{-1/4} G_t R_t^{-1/4}$$

(a) (5 pts) What do L_t and R_t represent geometrically? What structure are they capturing about the gradient sequence?

(b) (5 pts) Why are fractional powers (specifically $-1/4$) used instead of full inverses (-1)?

ANSWER:

(a) L_t and R_t represent **covariance matrices** of the gradient:

Left accumulator: $L_t = \sum_{s=1}^t G_s G_s^T \in \mathbb{R}^{d_{\text{out}} \times d_{\text{out}}}$

- Captures covariance structure in the **output space**
- $L_t \approx \mathbb{E}[GG^T]$ estimates how gradients covary across output dimensions

Right accumulator: $R_t = \sum_{s=1}^t G_s^T G_s \in \mathbb{R}^{d_{\text{in}} \times d_{\text{in}}}$

- Captures covariance structure in the **input space**
- $R_t \approx \mathbb{E}[G^T G]$ estimates how gradients covary across input dimensions

These capture the **curvature information** and identify which directions have historically had large vs. small gradients, allowing the optimizer to adapt to the geometry of the loss landscape.

(b) Fractional powers provide **preconditioning** rather than exact Newton steps:

Using full inverses $L_t^{-1} G_t R_t^{-1}$ would give:

$$(GG^T)^{-1} G (G^T G)^{-1}$$

This is too aggressive - it's a full Newton-like step that can be unstable, especially early in training when statistics are noisy.

Using $-1/4$ powers:

- Provides **partial whitening** (reduces but doesn't eliminate anisotropy)
- More **stable** numerically (matrices remain well-conditioned)
- Can be viewed as **interpolating** between SGD ($\alpha = 0$, no preconditioning) and Newton ($\alpha = 1$, full preconditioning)
- The $1/4 + 1/4 = 1/2$ total gives a balanced preconditioner that "square roots" the curvature

This is the Shampoo design choice: enough preconditioning to help, but not so much as to cause instability.

10 Question 10 (15 points)

Consider Shampoo **without accumulation**, so $L = GG^T$ and $R = G^T G$ at a single step. Let $G = U\Sigma V^T$ be the SVD with $\Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_r)$.

- (a) (5 pts) Express GG^T and $G^T G$ in terms of U , V , and Σ .
- (b) (6 pts) Compute $(GG^T)^{-1/4}$ and $(G^T G)^{-1/4}$ in terms of the SVD components.
- (c) (4 pts) Show that $(GG^T)^{-1/4}G(G^T G)^{-1/4} = UV^T$.

ANSWER:

- (a) Using the SVD $G = U\Sigma V^T$:

$$GG^T = (U\Sigma V^T)(V\Sigma^T U^T) = U\Sigma V^T V \Sigma U^T = U\Sigma^2 U^T$$

$$G^T G = (V\Sigma^T U^T)(U\Sigma V^T) = V\Sigma^T U^T U \Sigma V^T = V\Sigma^2 V^T$$

- (b) For a matrix of the form $A = Q\Lambda Q^T$ where Q is orthogonal:

$$A^\alpha = Q\Lambda^\alpha Q^T$$

Therefore:

$$(GG^T)^{-1/4} = (U\Sigma^2 U^T)^{-1/4} = U(\Sigma^2)^{-1/4} U^T = U\Sigma^{-1/2} U^T$$

$$(G^T G)^{-1/4} = (V\Sigma^2 V^T)^{-1/4} = V(\Sigma^2)^{-1/4} V^T = V\Sigma^{-1/2} V^T$$

- (c) Computing the full expression:

$$\begin{aligned} & (GG^T)^{-1/4} G (G^T G)^{-1/4} \\ &= (U\Sigma^{-1/2} U^T) \cdot (U\Sigma V^T) \cdot (V\Sigma^{-1/2} V^T) \\ &= U\Sigma^{-1/2} \cdot (U^T U) \cdot \Sigma V^T \cdot (V V^T) \cdot \Sigma^{-1/2} V^T \\ &= U\Sigma^{-1/2} \cdot I \cdot \Sigma \cdot I \cdot \Sigma^{-1/2} V^T \\ &= U \cdot (\Sigma^{-1/2} \cdot \Sigma \cdot \Sigma^{-1/2}) \cdot V^T \\ &= U \cdot I \cdot V^T = \boxed{UV^T} \end{aligned}$$

This shows Shampoo (without accumulation) produces exactly the semi-orthogonal update direction that comes from spectral-norm constrained optimization!

11 Remaining Questions 11-20

The exam contains 10 additional questions following the same format:

11. **Q11 (8 points):** Computational complexity - SVD cost vs Newton-Schulz cost vs Muon cost
12. **Q12 (12 points):** Newton-Schulz polynomial behavior and convergence analysis with $\sigma = 0.5$ and $\sigma = 1.5$
13. **Q13 (10 points):** Why odd polynomials preserve singular vectors (detailed proof)
14. **Q14 (8 points):** Normalization necessity for Newton-Schulz stability
15. **Q15 (12 points):** Practical vs classic polynomials (NanoGPT example), advantages and disadvantages
16. **Q16 (10 points):** Complete Muon algorithm in pseudocode and momentum's role
17. **Q17 (8 points):** True/False questions (4 parts) on key concepts
18. **Q18 (12 points):** P scaling in practice - width transfer from 512 to 4096
19. **Q19 (10 points):** Norm relationships and conservativeness proofs
20. **Q20 (15 points):** Synthesis narrative connecting all concepts

Each question includes:

- Detailed problem statement
- Horizontal line separator
- Complete solution with step-by-step work
- Explanations connecting to broader themes